

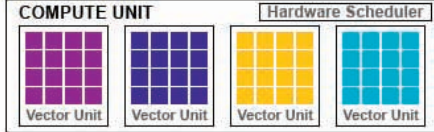
NVIDIA tech in automobiles

NVIDIA Tegra 3 platform will power in-car digital systems in Audi A7, Tesla Model S and Lamborghini Aventador

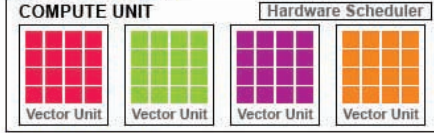
Transformer ICS update

According to ASUS North America's reply to a user query, the Ice Cream Sandwich update will arrive on the ASUS Transformer tablet by mid-February

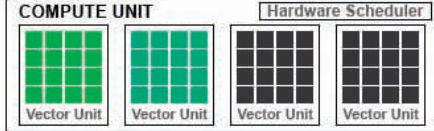
First Clock Cycle:



Second Clock Cycle:



Third Clock Cycle:



Dependency handling in CU

dependent wavefronts in the next clock cycle. This ensures that unlike VLIW 4 setup, none of the ALUs go unused.

The vector units are just one part of the compute unit. The CU also has the Scalar Unit which handles the pointer arithmetic and branching code, leaving the vector unit to just concentrate on the wavefronts that are pixel heavy. A scalar unit handles simple integer operations, if/else conditional branches, etc., saving up on the vector unit time. So each unit is optimised to perform tasks that it's best at. There are four Texture filter units on each CU block.

Four compute units share an L1 cache which comprises a 16KB Instruction cache and 32KB Scalar data cache. This ensures smooth communication between multiple compute units. So if a thread in CU1 requires some data and is dependent on another thread in CU2 or CU3 for instance, then that data can be shared over the L1 cache.

The new GPU

Compute unit, even though it's a minor processor in itself, is just a building block of the whole GPU. Looking at the overall diagram of the GPU alongside, you'll come across blocks such as an Asynchronous Compute Engine Command processor, Graphics command processor, Primitive Pipe, Pixel pipes and Raster Operators.

Asynchronous Compute Engine's (ACE) main aim is to accept work and then distribute it to the compute units for further processing. There can be multiple ACEs within a GPU and they have the ability to pri-

oritise or reprioritise tasks based on dependencies so that they're completed in a different order than what they were received in.

For graphics, there is the Graphics Command Processor, below which you have the multiple Primitive Pipes. The pipes contain the geometry assembler, Gen 9 Tessellator and Vertex assembler. The tessellation units feature off-chip buffering. This means that the geometry data from tessellated workloads can be stored in the DRAM only if the on-chip cache becomes full.

Once the compute units are done with the compute and graphical workload received from the ACE and GCP, the output is then sent to the Pixel pipelines. These contain the Raster operators which write the pixel on the output buffer i.e. the L2 cache of the GPU.

Memory and cache

Each compute unit has its own L1 cache that is divided into data, instruction and load/store. The GPU, on the other hand, has L2 cache which is fully coherent, maintaining data consistency. There is a 64-bit interface between L1 and L2 cache and with scaling

in the number of compute units, there will be an increased bandwidth between these caches. Moreover, the L1 cache is read/write which makes the CU even more optimised for future workloads.

Also this architecture supports virtualisation of memory with the CPU. In previous generation cards, if the data set was too large to be accommodated on the onboard cache or VRAM, then there had to be a provision for separate memory virtualisation system, from the CPU's virtual memory.

But with x86-64 virtual memory support, the GPU can share the CPU's virtual memory thus improving data handling. So things like Megatextures – which are single large texture spaces for static terrains – currently loaded from the system hard drive (as they can't be accommodated in the graphics cards VRAM), can be stored in the virtual memory with Graphics Core Next, which ensures quicker load times.

So that's about the new architecture which will be seen in Radeon HD 79xx series of cards only. Refer to the block diagram below. ■



The new architecture